

Department of Computer and Software Engineering**SE: Machine Learning****Course Instructor:****Date:****Day:****Semester:****Lab 9: Imbalanced Fraud Detection**

Lab Title	CLO	BT	PLO	Weightage
Imbalanced Fraud Detection F1 vs Accuracy & Cost-Sensitive Learning				

Name	Roll Number	Report /10	Viva /5	Total /15

Checked on: _____

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Objectives

- Understand challenges in imbalanced datasets
- Compare Accuracy vs F1-score
- Apply sampling techniques (up/down/SMOTE/SMOTEEN)
- Implement cost-sensitive logistic regression

Problem Statement

You are building a fraud detection system where fraudulent transactions are rare compared to normal ones.

You will analyze how class imbalance affects model performance and explore different techniques to handle imbalance.

Part A: Dataset Exploration

Task 1: Analyze Dataset 1 (60:40)

Load the provided dataset with a 60:40 class distribution (normal vs fraud). Visualize the dataset and observe how the two classes are distributed in the feature space.

Task 2: Analyze Dataset 2 (90:10)

Load the provided dataset with a higher imbalance (90:10). Visualize and compare it with Dataset 1. Observe how class imbalance affects separability.

Part B: Logistic Regression and Metrics

Task 3: Train Logistic Regression

Train a logistic regression model using cross-entropy loss:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Task 4: Evaluate Accuracy and F1 Score

Compute:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}}$$

Compare results for both datasets.

Part C: Sampling Techniques

Task 5: Apply SMOTE

Apply: - SMOTE (synthetic minority oversampling)
Observe the effect on model performance.

Task 6: Apply SMOTEEN

Apply: - SMOTEEN (SMOTE + cleaning)
Compare results with previous methods.

Part D: Cost-Sensitive Learning

Task 7: Weighted Loss Function

Modify the loss function to penalize misclassification of fraud more:

$$L = - \sum_{i=1}^N [w_1 y_i \log(p_i) + w_0 (1 - y_i) \log(1 - p_i)]$$

Experiment with different class weights and observe changes.

Discussion Questions

- Why can accuracy be misleading in imbalanced datasets?
- Which method improved F1-score the most?
- How does cost-sensitive learning compare to sampling methods?